

1 MAKING INDIA THE HUB FOR QUANTUM CYBERSECURITY

QNu Labs, pronounced as Q-New, is India's only quantum-tech cybersecurity company. It is trying to reimagine cybersecurity in a futuristic way, build it here in India, and then take it to the whole world



Sunil Gupta, Co-founder and CEO of QNu Labs

Encryption today is nothing more than a mathematical problem, and just like you can solve mathematical problems, mathematics based encryption systems can be hacked. QNu Labs has hence redefined cybersecurity and built its foundation on the quantum physics engine instead of relying on mathematics. "Our products are an integration of three things—elec-

tronics, photonics (photon-based electronics), and quantum physics," says Sunil Gupta, Co-founder and CEO of QNu Labs.

Their Quantum Cryptography platform is based on four engines—Tropos, Armos, Hodos, and Qosmos. While Hodos and Qosmos are software engines, Tropos and Armos are hardware based and employ some specialised optoelectronic compo-

nents. For example, in Tropos—the quantum random number generator (QRNG)—photons generated from a laser are eventually used to generate random numbers. Since this is a hardware based process with no human intervention, these random numbers are truly random!

QNu Labs has also built several packaged solutions on top of their quantum engines. Some of them are Quantum Entropy-as-a-Service, Quantum Secure Email, Quantum end-to-end encrypted chat application called Qverse, and Quantum VPN Tunnel. Moreover, their random number generator can be bought as an appliance, used as a cloud service via AWS, and is also available as a small chip for IoT devices. "We think that for the new technologies—5G, AI, ML, IoT, blockchains, etc—quantum cryptography has very interesting applications," says Gupta.

QNu Labs has huge plans for expansion. The company has already put India on the quantum map of the world. They strongly believe that India is the right place to build such technology and deploy it. Talking about QNu Labs' expansion strategy, Gupta says, "India is a digital economy with so many people using digital solutions. Indian philosophy